

Sequential Circuits

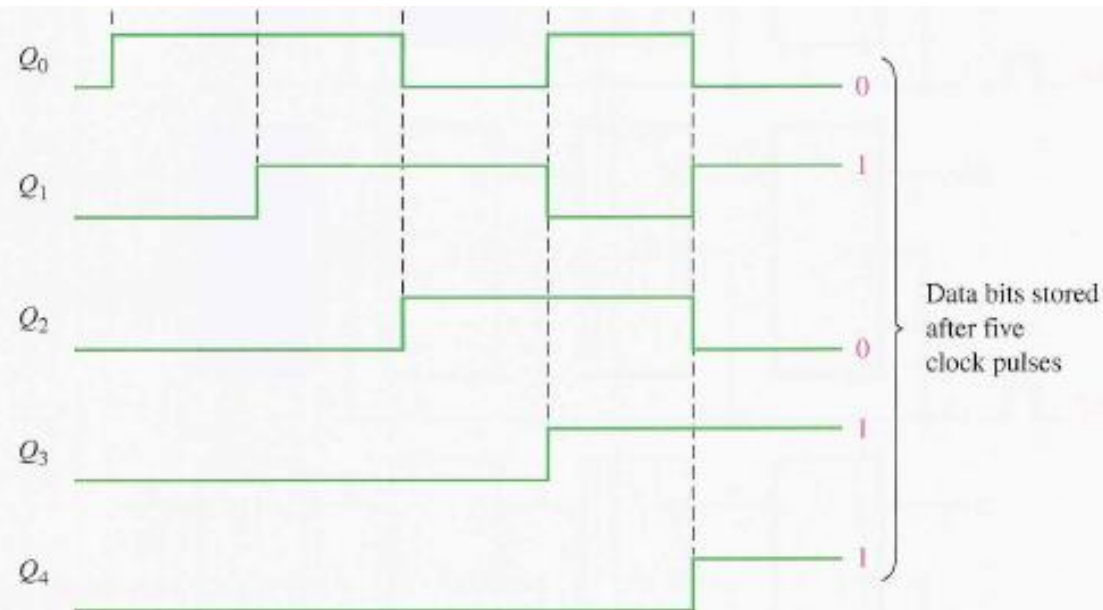
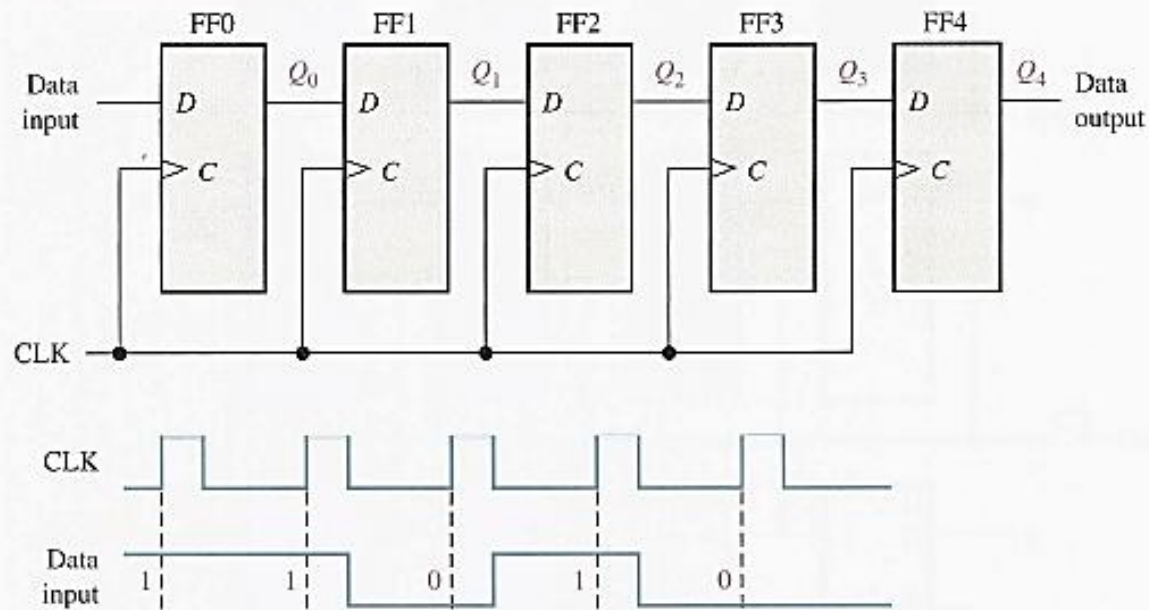
Tutorial 1

Registers

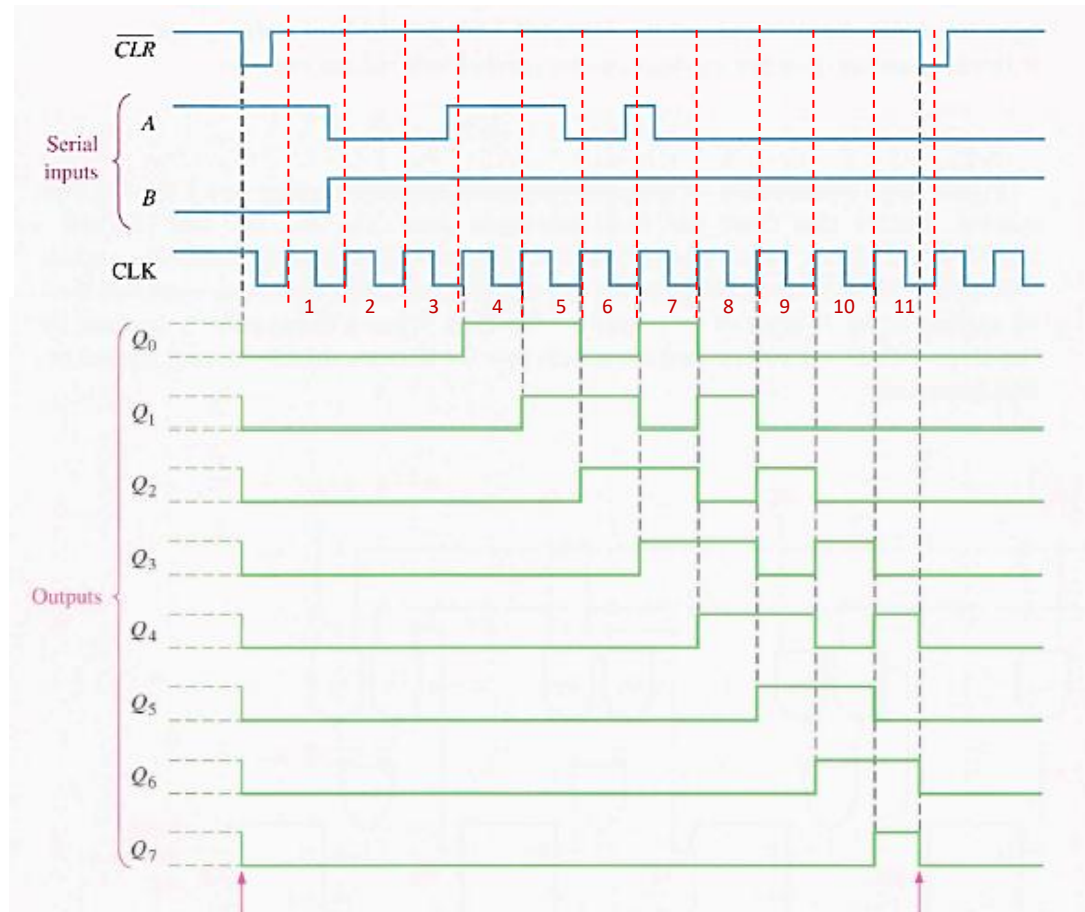
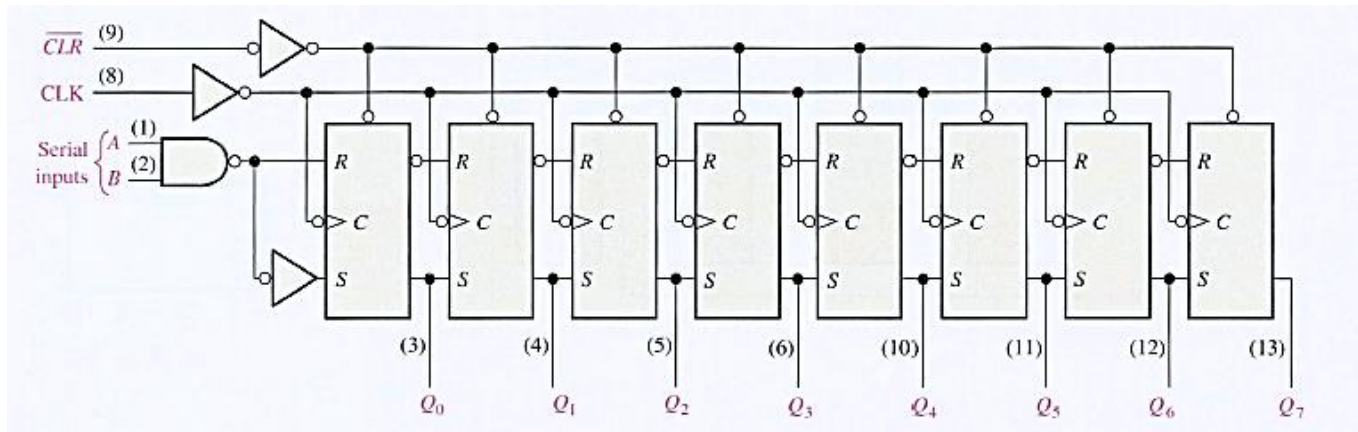
15-11-2023

Q1/

Show the states of the 5-bit register in Figure for the specified data input and clock waveforms. Assume that the register is initially cleared (all 0s).



Q2/ Show the output states for the 8 bits serial in parallel out shift register



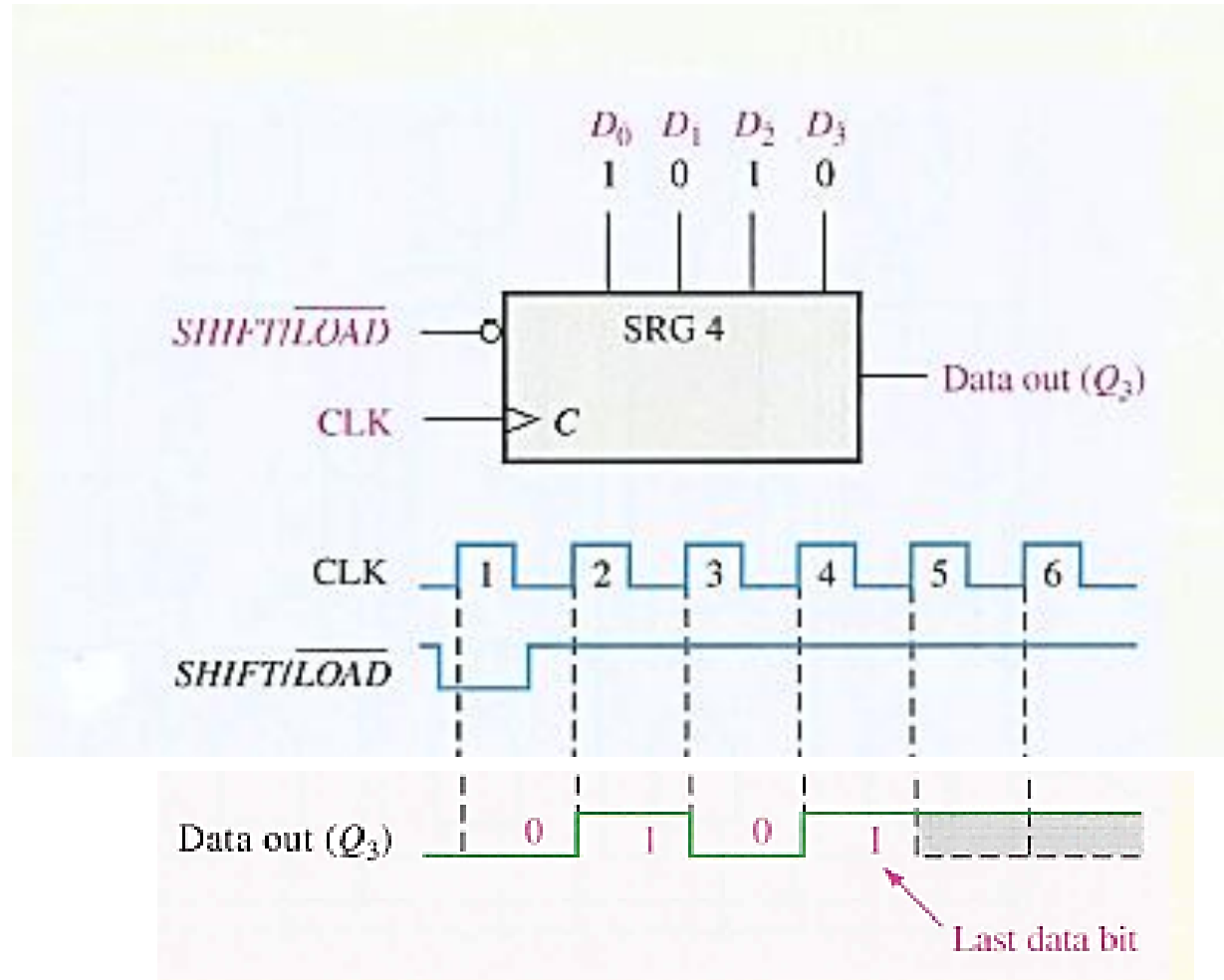
Q3/

The bit sequence 1101 is serially entered (right-most bit first) into a 4-bit parallel out shift register that is initially clear. What are the Q outputs after two clock pulses?

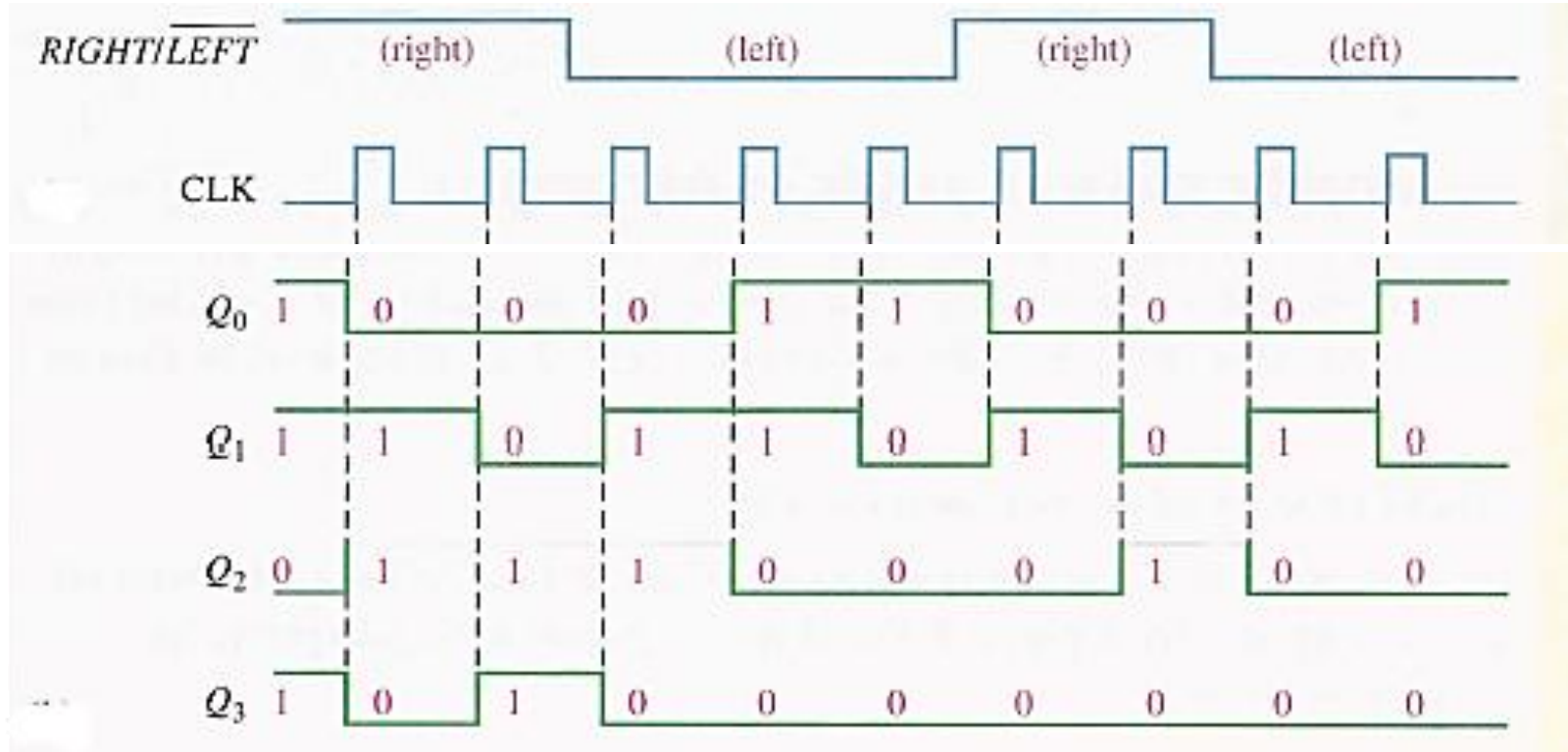
Sol: 0100

Q4/

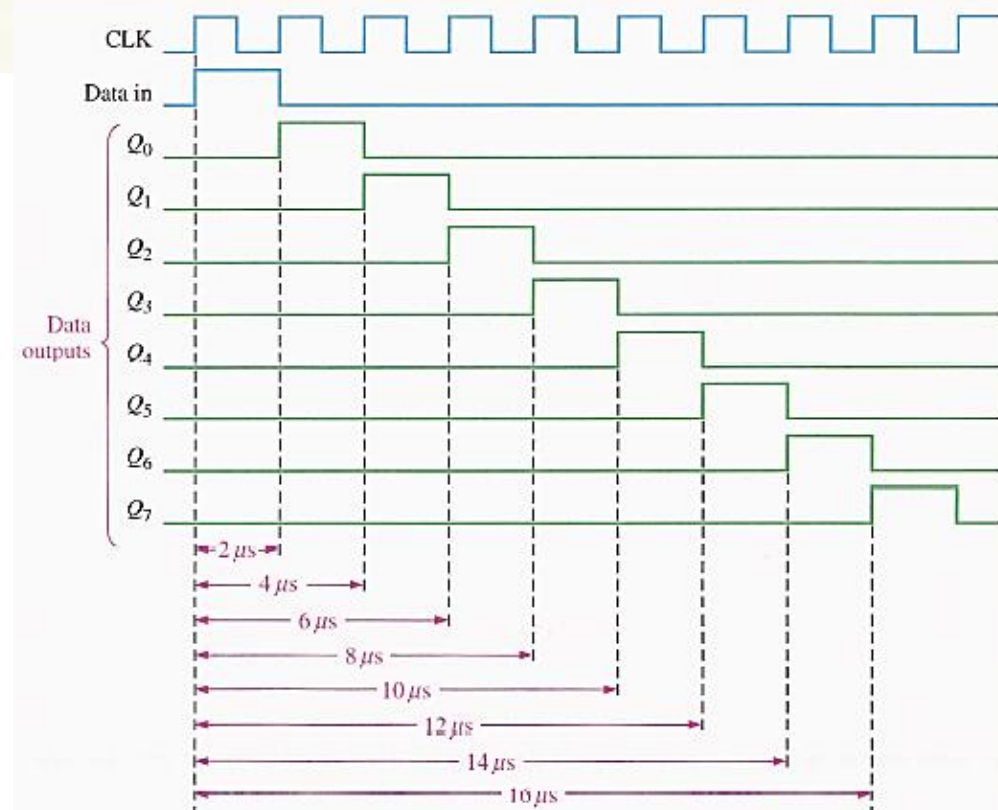
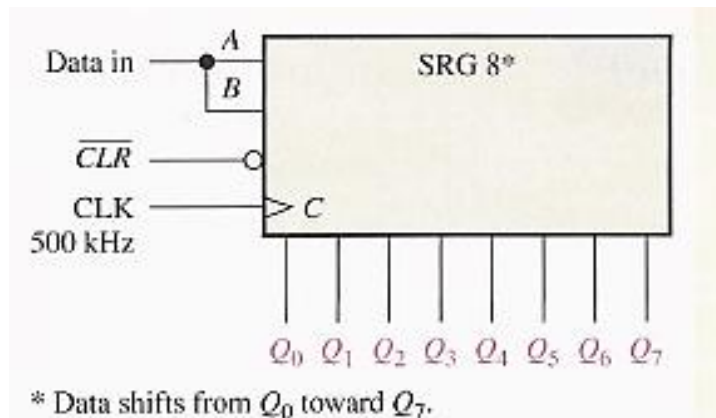
Show the data output waveform for the 4 bit parallel load register shown



Q5/ For a bidirectional shift register draw the output waveform. Assume $Q_0 = 1$, $Q_1 = 1$, $Q_2 = 0$, $Q_3 = 1$



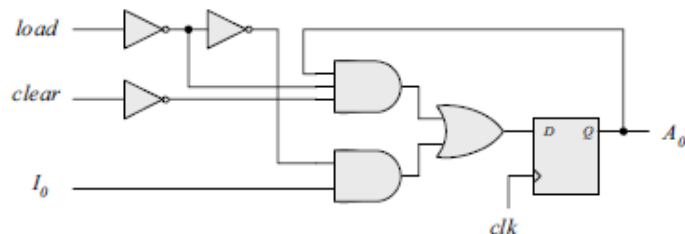
Q6/ Determine the amount of time delay between the serial input and each output in Figure . Show a timing diagram to illustrate.



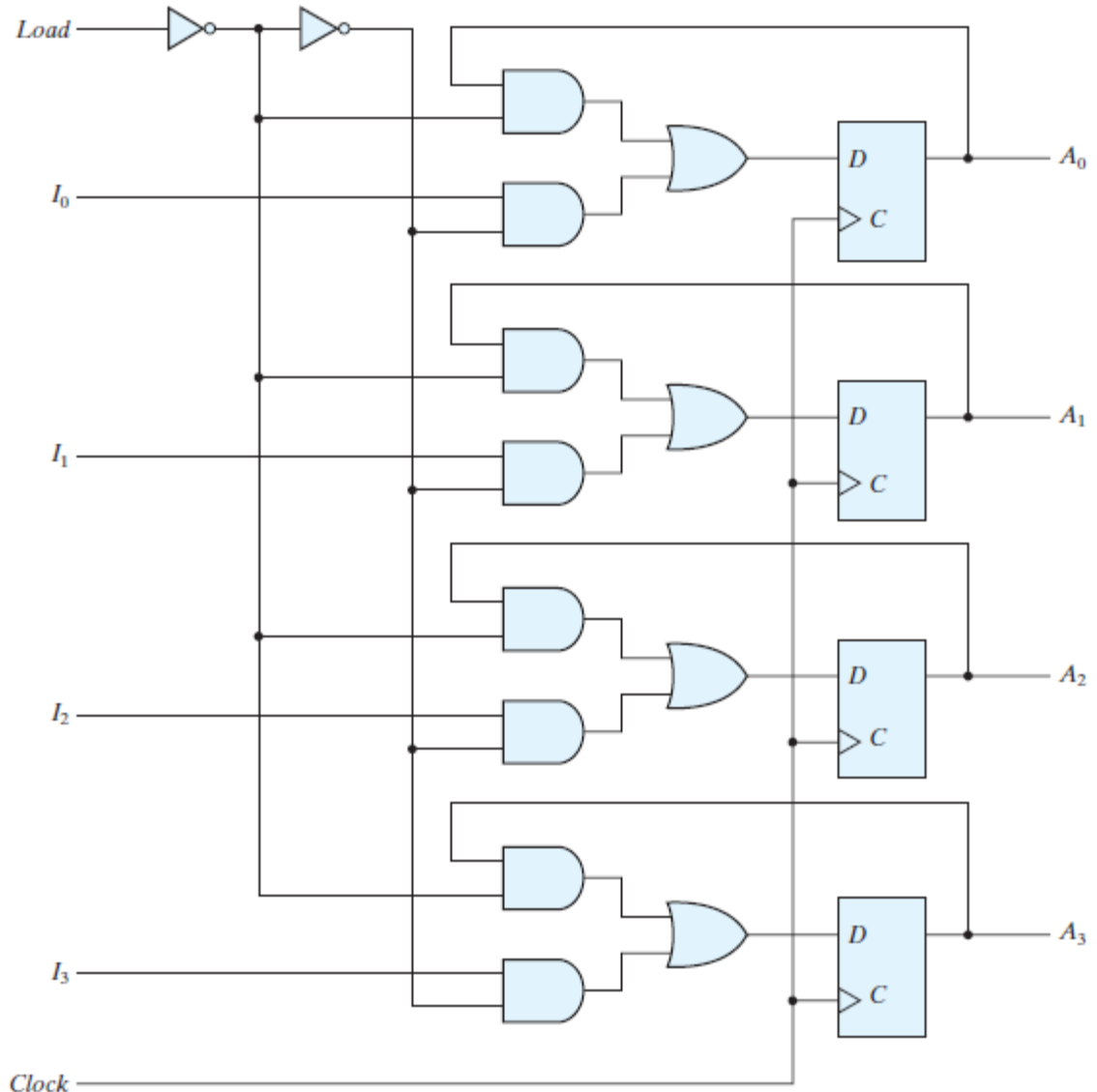
Q7/

6.2) Include a synchronous clear input to the register of Fig. 6.2 . The modified register will have a parallel load capability and a synchronous clear capability. The register is cleared synchronously when the clock goes through a positive transition and the clear input is equal to 1.

Sol:



<i>Load</i>	<i>Clear</i>	<i>D</i>	<i>Operation</i>
0	0	<i>A₀</i>	No change
0	1	0	Clear to 0
1	x	<i>I₀</i>	Load input



Q8/ **6.4)** The contents of a four-bit register is initially 0110. The register is shifted six times to the right with the serial input being 1011100. What is the content of the register after each shift?

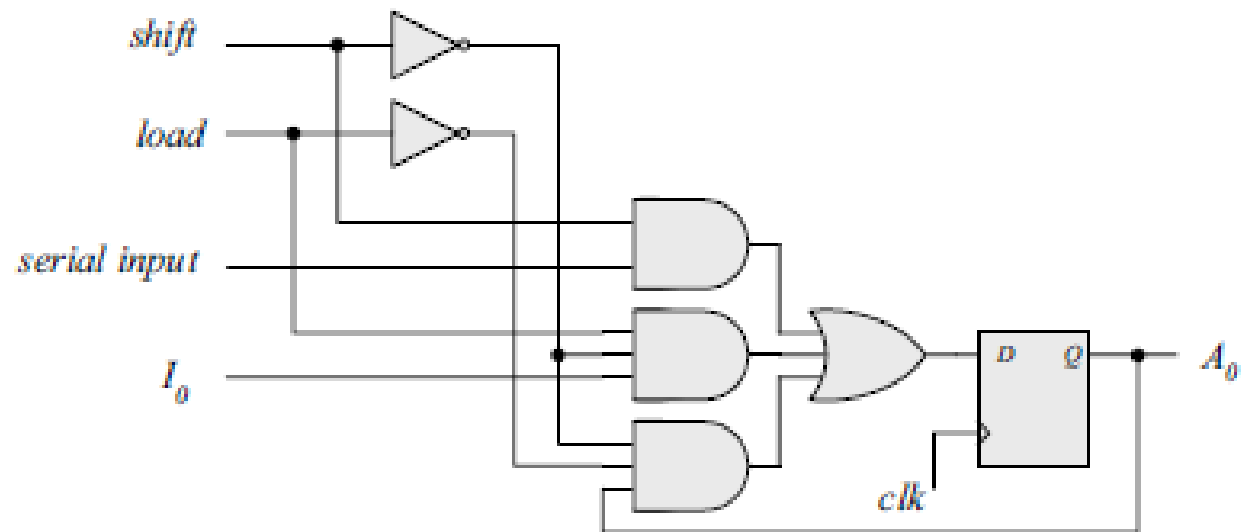
Sol:

Last output is : 0111

Q9/

6.6 Design a four-bit shift register with parallel load using D flip-flops. There are two control inputs: *shift* and *load*. When *shift* = 1, the content of the register is shifted by one position. New data are transferred into the register when *load* = 1 and *shift* = 0. If both control inputs are equal to 0, the content of the register does not change.

First stage of register:

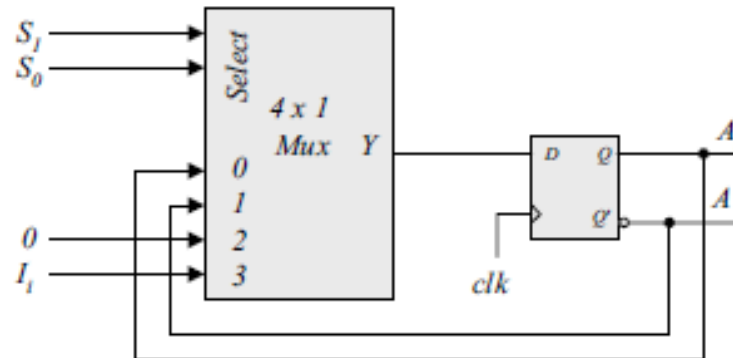


Q10/

6.7) Draw the logic diagram of a four-bit register with four D flip-flops and four 4×1 multiplexers with mode selection inputs s_1 and s_0 . The register operates according to the following function table.

s_1	s_0	Register Operation
0	0	No change
1	0	Complement the four outputs
0	1	Clear register to 0 (synchronous with the clock)
1	1	Load parallel data

First stage of register:



The serial adder of Fig. 6.6 uses two four-bit registers. Register *A* holds the binary number 0101 and register *B* holds 0111. The carry flip-flop is initially reset to 0. List the binary values in register *A* and the carry flip-flop after each shift.